Introduction to Economics

Class 13 — Monopoly; oligopoly and the prisoner's dilemma

Pavel Kocourek

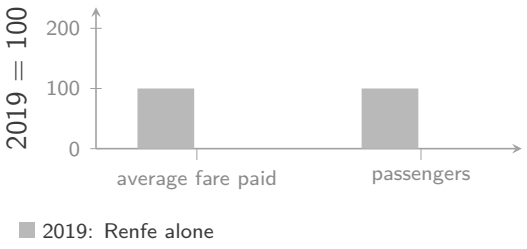
Universitat de Barcelona · Group A6

9 November 2026

Plan for today

- Monopoly: the price and quantity a single seller picks
- What a monopoly costs the rest of us
- A price for each buyer; what governments do
- Oligopoly: two sellers and a deal between them
- More sellers; when a deal can hold
- Practice: today's moves; a tax, a tariff, an external cost

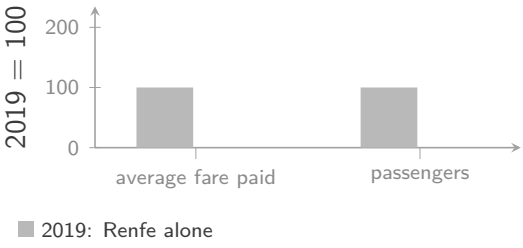
Madrid–Barcelona by fast train: three companies instead of one — fares down 32 %, passengers doubled



until May 2021: Renfe only

Comisión Nacional de los Mercados y la Competencia (CNMC), Informe anual del sector ferroviario 2025 (INF/DTSP/039/26, 30 June 2026), p. 39, table 23: Madrid–Barcelona, 2025 vs 2019; fare = nominal ticket revenue per station-to-station passenger (52.89 euros in 2025), 8.7 million passengers in 2025; entry dates: table A2, p. 56.

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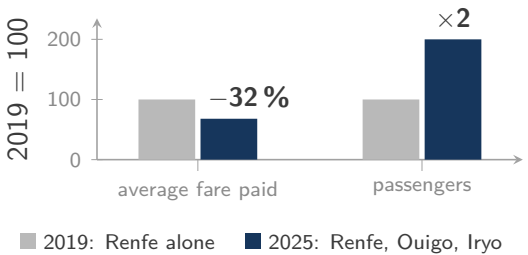
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Why might more sellers push the fare down?

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From the second hundred on, each adds less to revenue than its price: the price falls for all

highest price for all (euros)	13	12	11	10	9	8	7	6
rides a day	100	200	300	400	500	600	700	800

Marginal revenue: what one more ride adds to revenue
(here: per ride, for each extra hundred)

Monopoly: the only seller of a good with no close substitute; it faces the market's demand (class 12, slide 20).

A price taker's extra bag adds just the price: marginal revenue (MR) = P = 4 (class 12, slides 6 and 8).

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marginal revenue (euros a ride)	13	11	9	7	5	3	1	−1

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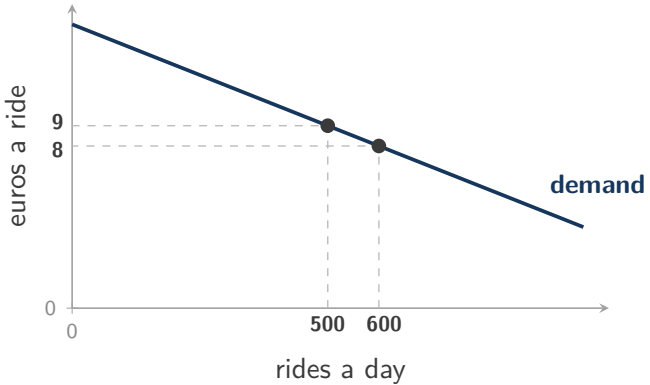
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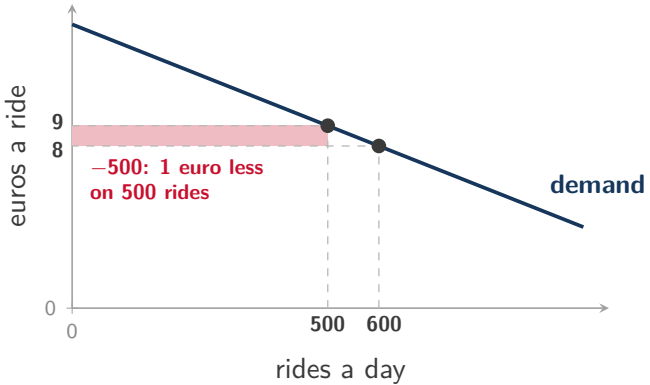
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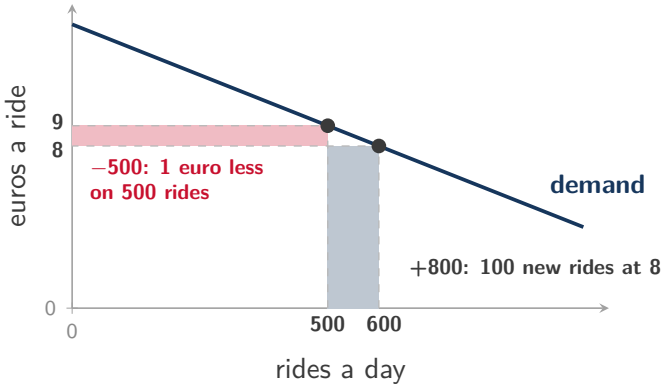
From 500 to 600 rides: 800 more from new riders, 500 less from the old ones



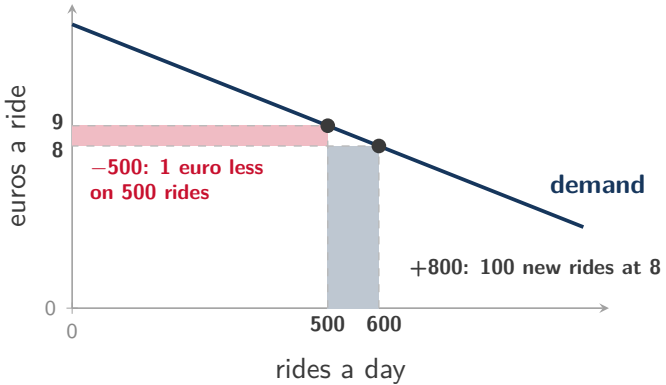
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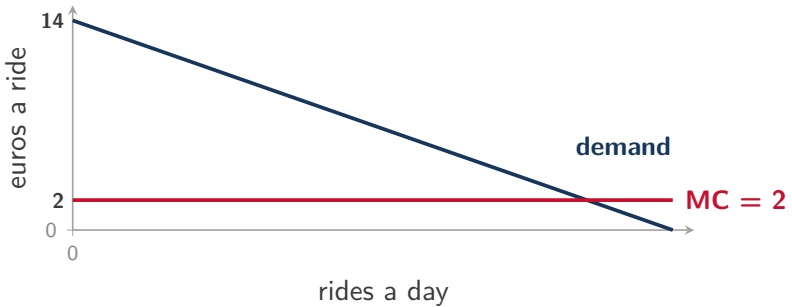
From 500 to 600 rides: 800 more from new riders, 500 less from the old ones



Marginal revenue: $800 - 500 = 300$ for 100 rides — 3 a ride, below the price of 8.

Sell every hundred whose marginal revenue beats the 2-euro cost: 600 rides at 8 euros

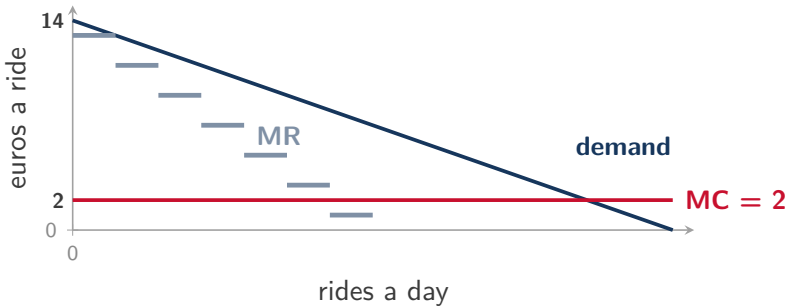
Each extra ride costs 2 euros (energy, wear); fixed costs are paid whatever it sells.



Marginal cost (MC): what one more ride adds to total cost (class 11, slide 10). A price taker produces while MC is below the price (class 12, slide 8).

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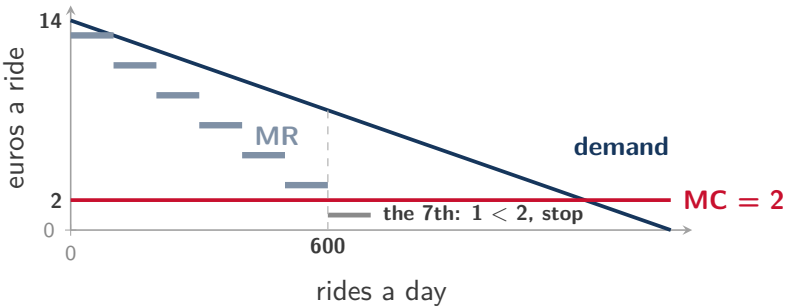
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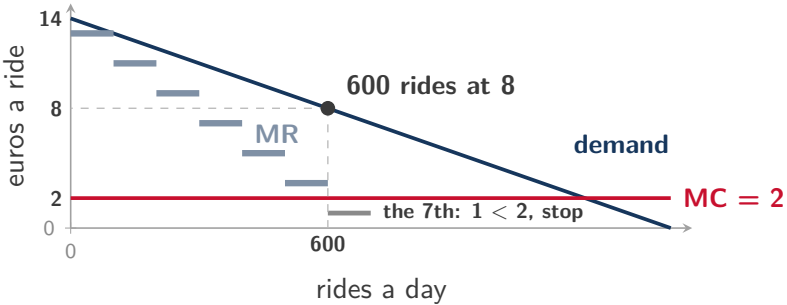
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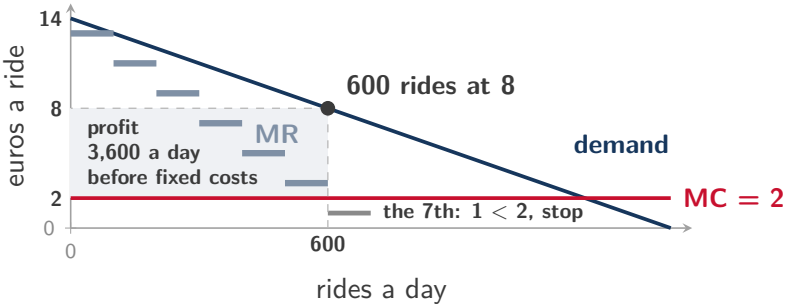
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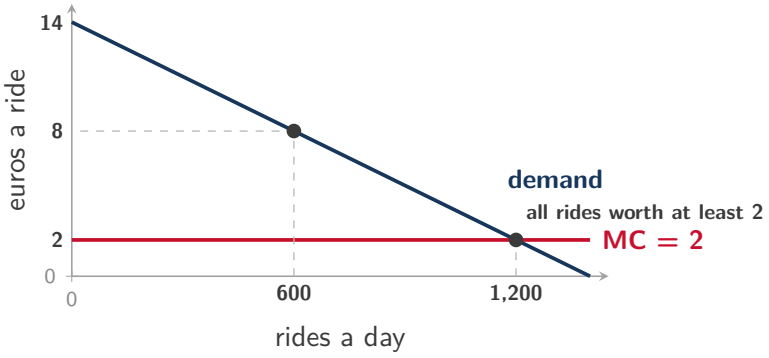
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MR falls below MC after 600; demand sets the price: 8, far above MC; profit $(8 - 2) \times 600$.

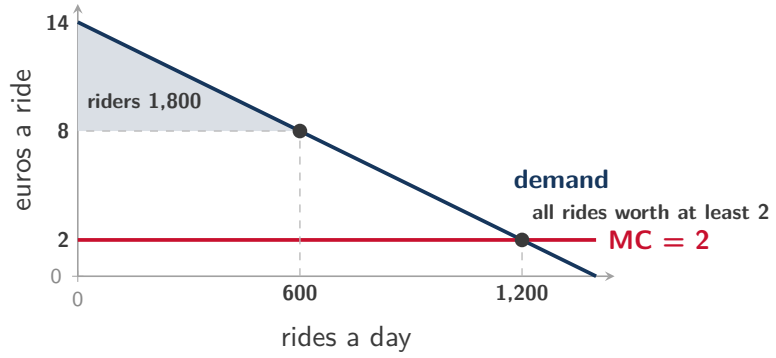
Marginal cost (MC): what one more ride adds to total cost (class 11, slide 10). A price taker produces while MC is below the price (class 12, slide 8).

Rides worth more than their 2-euro cost go unsold: 1,800 euros a day lost



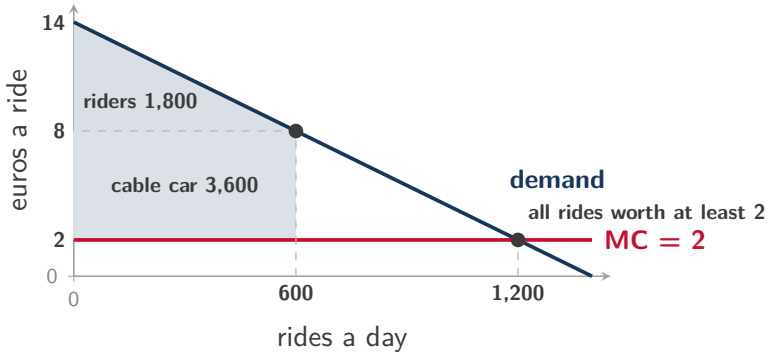
Consumer surplus: value minus price (class 8, slide 6) · deadweight loss: surplus lost when trades worth more than their cost do not happen (class 8, slide 17).

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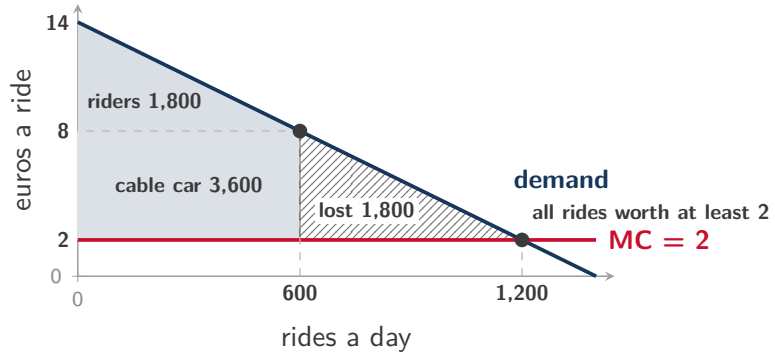
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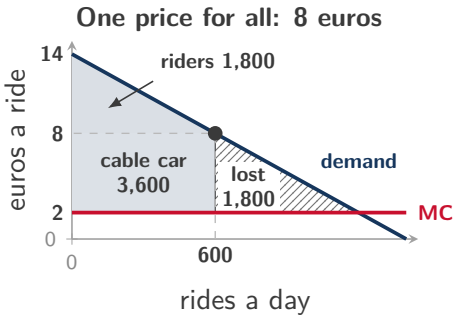
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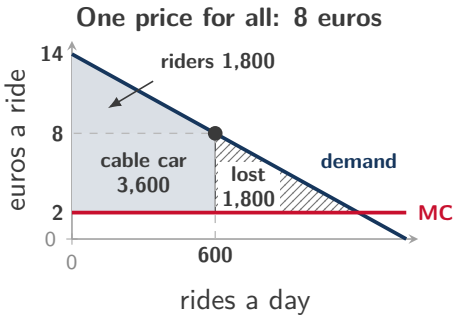
The 3,600 is a transfer from riders; the 1,800 is lost to everyone.

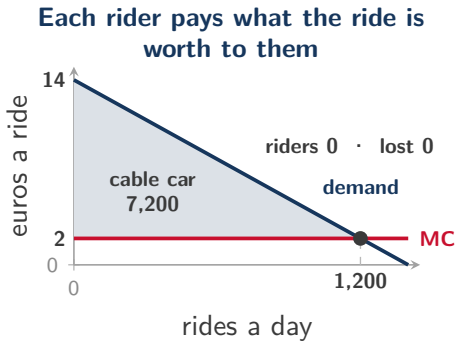
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A price for each rider: all 1,200 rides sold, nothing lost — and the cable car keeps all 7,200

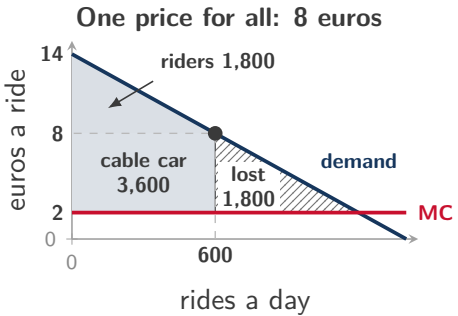


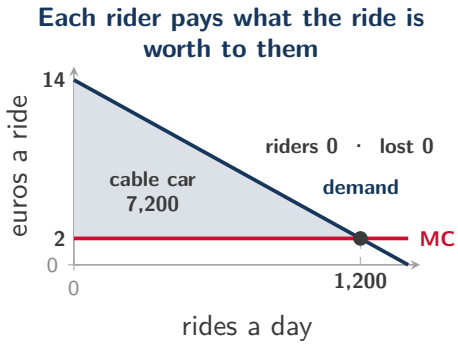
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Price discrimination: one good, different prices for different buyers.
In practice by group: a residents' card, youth fares, booking early.

Four answers to a monopoly: competition law, a price cap, public ownership — or nothing

- **Competition law:** stop mergers, fine price-fixing

Natural monopoly: one firm serves the whole market most cheaply (class 12, slide 21).

In Spain: the CNMC (Comisión Nacional de los Mercados y la Competencia). Reading Ch. 11, pp. 258–261.

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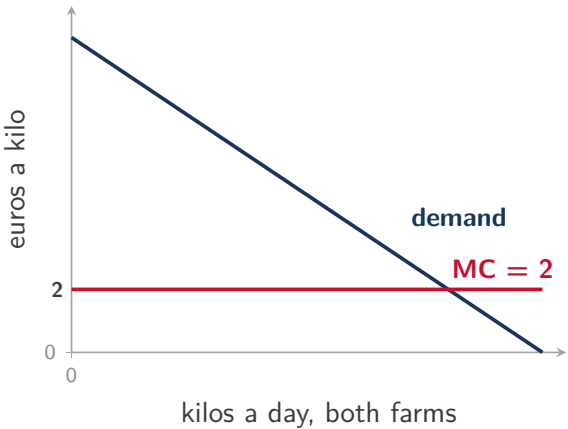
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- **Doing nothing:** when every cure costs more

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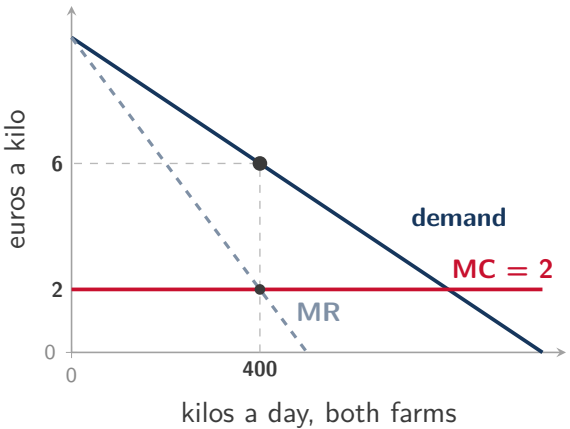
Two mussel farms share one bay: acting as one monopoly, they would sell 400 kilos at 6 euros



Perla and Onada: the only two farms; each kilo costs 2 to farm.

Oligopoly: a market with a few sellers of similar goods

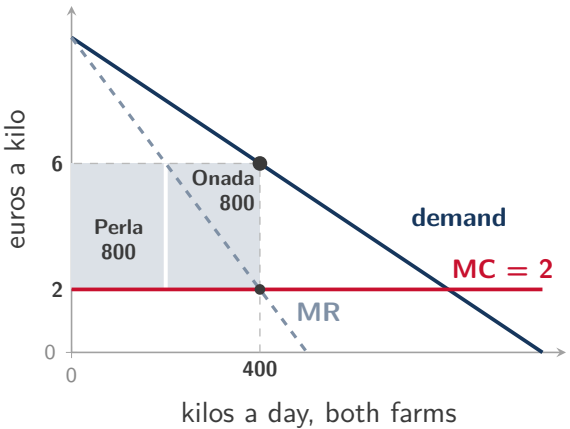
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Oligopoly: a market with a few sellers of similar goods

Cartel: sellers who agree to act as one monopoly: here 200 kilos each

Selling 250 kilos instead of 200 pays each farm more — whatever the other sells

Each farm sells 200 kilos (the deal) or 250.
Price: 400 kilos $\rightarrow$ 6.00 · 450 $\rightarrow$ 5.50 · 500 $\rightarrow$ 5.00.

euros a day, Perla's first

		Onada	
		200 kilos	250 kilos
Perla	200 kilos	Perla 800, Onada 800	
	250 kilos		

profit = (price − 2) × own kilos

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		Onada	
		200 kilos	250 kilos
Perla	200 kilos	Perla 800, Onada 800	Perla 700, Onada 875
	250 kilos	Perla 875, Onada 700	

$$\text{profit} = (\text{price} - 2) \times \text{own kilos}$$

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Perla	200 kilos	Perla 800, Onada 800	Perla 700, Onada 875
	250 kilos	Perla 875, Onada 700	Perla 750, Onada 750

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	250 kilos	Perla 875, Onada 700	Perla 750, Onada 750

$$\text{profit} = (\text{price} - 2) \times \text{own kilos}$$

If Onada keeps the deal, Perla earns 875 by breaking it, not 800.

Both sell 250 kilos: neither gains by changing alone — yet both earn less than with the deal

euros a day, Perla's first

		Onada	
		200 kilos	250 kilos
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	250 kilos	Perla 875, Onada 700	Perla 750, Onada 750

Dominant strategy: best whatever the other does

Both sell 250 kilos: neither gains by changing alone — yet both earn less than with the deal

euros a day, Perla's first

		Onada	
		200 kilos	250 kilos
Perla	200 kilos	Perla 800, Onada 800	Perla 700, Onada (875)
	250 kilos	Perla (875), Onada 700	Perla (750), Onada (750)

Dominant strategy: best whatever the other does

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euros a day, Perla's first

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		200 kilos	250 kilos
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Dominant strategy: best whatever the other does

Nash equilibrium: each does its best, given what the other does

Both sell 250 kilos: neither gains by changing alone — yet both earn less than with the deal

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Dominant strategy: best whatever the other does

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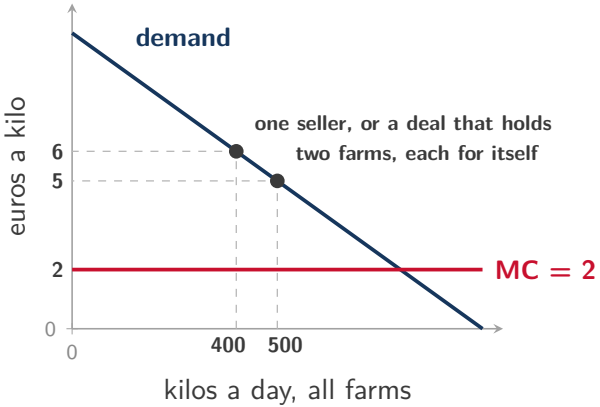
Prisoner's dilemma: each does its best, and both are worse off than if both had cooperated

More sellers, a price closer to cost: 6, then 5, then 2 euros a kilo



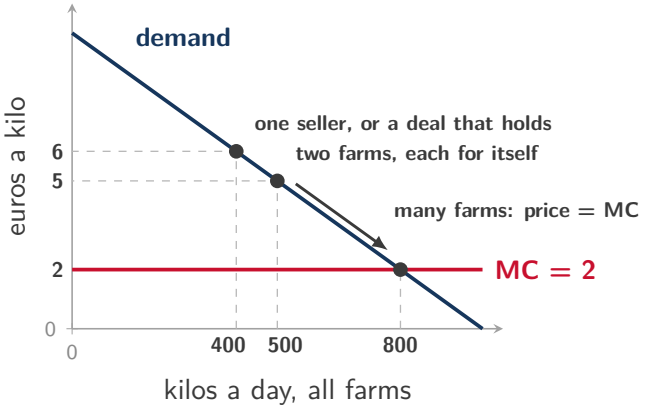
With free entry, the price falls to the lowest average total cost (class 12, slide 18).

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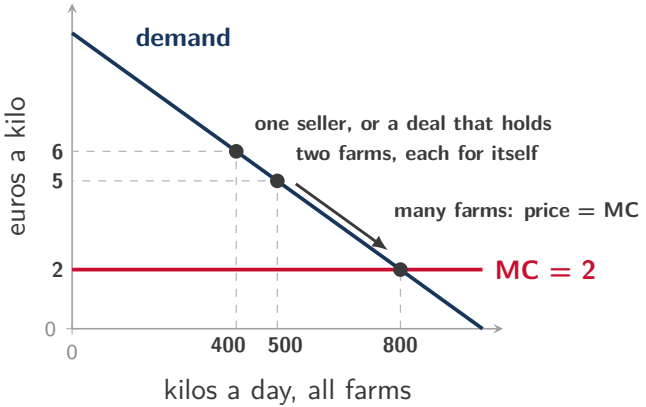
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Madrid–Barcelona

one company → three
average fare: −32 %
passengers: ×2

2019 → 2025
(CNMC, slide 3)

With free entry, the price falls to the lowest average total cost (class 12, slide 18).

Meeting every day, the deal can hold: a cheater gains 75 once, then loses 50 every day

Perla's profit so far, euros

	day 1	day 2	day 3
Keeps the deal (200 kilos)	800	1,600	2,400

Meeting every day, the deal can hold: a cheater gains 75 once, then loses 50 every day

Perla's profit so far, euros

	day 1	day 2	day 3
Keeps the deal (200 kilos)	800	1,600	2,400
Breaks it on day 1; both sell 250 from day 2	875	1,625	

Meeting every day, the deal can hold: a cheater gains 75 once, then loses 50 every day

Perla's profit so far, euros

	day 1	day 2	**day 3**
Keeps the deal (200 kilos)	800	1,600	**2,400**
Breaks it on day 1; both sell 250 from day 2	875	1,625	**2,375**

Meeting every day, the deal can hold: a cheater gains 75 once, then loses 50 every day

Perla's profit so far, euros

	day 1	day 2	**day 3**
Keeps the deal (200 kilos)	800	1,600	**2,400**
Breaks it on day 1; both sell 250 from day 2	875	1,625	**2,375**

Tit-for-tat: start with the deal, then do what the other did yesterday.

In Spain, rivals' agreements on prices or quantities are banned: Ley 15/2007, art. 1.

A monopoly sells while marginal revenue beats marginal cost and reads its price off the demand curve.

Its price is above marginal cost: rides worth more than their cost go unsold.

Each member of a cartel gains by breaking it: a prisoner's dilemma.
More sellers push the price toward marginal cost.

Break

Back at 11:00

Multiple choice, two minutes alone: the monopoly's price

A1 [4 points]

One firm owns the town's only fibre network. Each extra household it connects costs it 10 euros a month. The table gives the highest price at which it can connect each number of households, one price for all. Which price gives the firm the largest profit?

price (euros a month)	60	50	40	30	20	10
households (hundreds)	1	2	3	4	5	6

A 20 euros a month
B 40 euros a month
C 10 euros a month
D 30 euros a month

Answer A1

A B C D

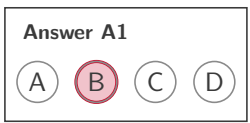
Multiple choice, two minutes alone: the monopoly's price

A1 [4 points]

One firm owns the town's only fibre network. Each extra household it connects costs it 10 euros a month. The table gives the highest price at which it can connect each number of households, one price for all. Which price gives the firm the largest profit?

price (euros a month)	60	50	40	30	20	10
households (hundreds)	1	2	3	4	5	6

A 20 euros a month
B 40 euros a month
C 10 euros a month
D 30 euros a month



B: the 3rd hundred adds 20 a household, the 4th only 0, below the cost of 10: 300 households at 40 euros

Multiple choice, two minutes alone: open on Sundays?

A2 [4 points]

A town's only two supermarkets, Fresc and Rodó, each choose whether to open on Sundays. Profit in thousand euros a month, Fresc's first:

		Rodó: closed	Rodó: open
Fresc	closed	Fresc 12, Rodó 10	Fresc 8, Rodó 13
	open	Fresc 15, Rodó 6	Fresc 9, Rodó 8

Which outcome is the Nash equilibrium?

A Fresc: closed; Rodó: closed.
B Fresc: open; Rodó: closed.
C Fresc: closed; Rodó: open.
D Fresc: open; Rodó: open.

Answer A2

A B C D

Multiple choice, two minutes alone: open on Sundays?

A2 [4 points]

A town's only two supermarkets, Fresc and Rodó, each choose whether to open on Sundays. Profit in thousand euros a month, Fresc's first:

		Rodó	
		closed	open
Fresc	closed	Fresc 12, Rodó 10	Fresc 8, Rodó 13
	open	Fresc 15, Rodó 6	Fresc 9, Rodó 8

Which outcome is the Nash equilibrium?

A Fresc: closed; Rodó: closed.
B Fresc: open; Rodó: closed.
C Fresc: closed; Rodó: open.
D Fresc: open; Rodó: open.

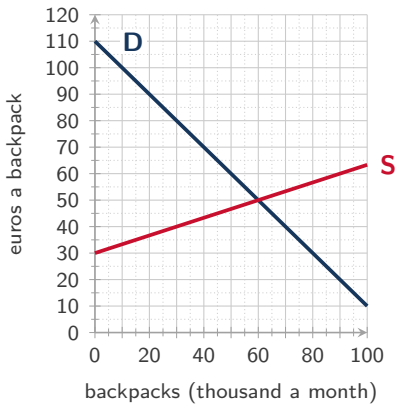
Answer A2

A B C D

D: opening is best for each, whatever the other does: Fresc 15 $>$ 12 and 9 $>$ 8; Rodó 13 $>$ 10 and 8 $>$ 6

A tax on backpacks: the line and the prices — three minutes alone

The grid shows the market for backpacks: demand D and supply S. The state puts a tax of 40 euros on every backpack sold; the sellers pay it to the state.



[5 points; drawing 2; boxes as marked]

(a) Draw the supply curve with the tax: at every quantity 40 euros above S. Label it S + tax.

(b) With the tax (quantity in thousand, prices in euros):

Quantity sold [1]

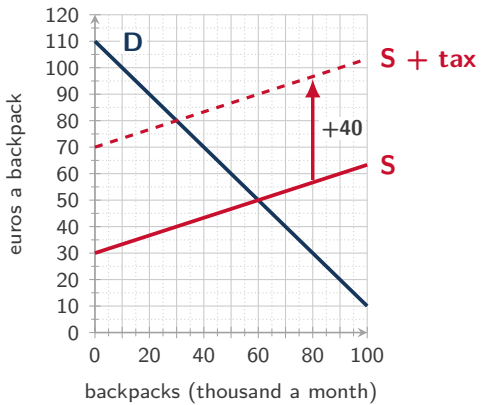
Buyers pay (euros a backpack) [1]

Sellers keep (euros a backpack) [1]

Draw on the grid and label every line you draw; numbers in the boxes, working beside them.

A tax on backpacks: the line and the prices — three minutes alone

The grid shows the market for backpacks: demand D and supply S. The state puts a tax of 40 euros on every backpack sold; the sellers pay it to the state.



[5 points; drawing 2; boxes as marked]

(a) Draw the supply curve with the tax: at every quantity 40 euros above S. Label it S + tax.

(b) With the tax (quantity in thousand, prices in euros):

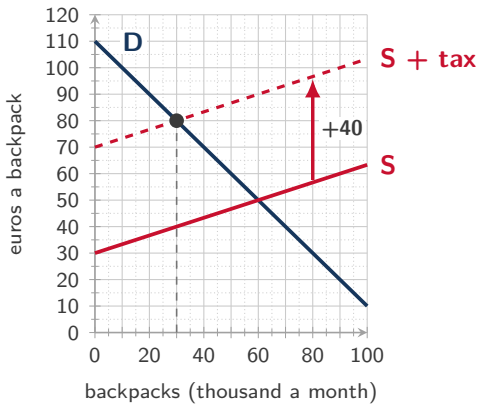
Quantity sold [1] Buyers pay (euros a backpack) [1]

Sellers keep (euros a backpack) [1]

Draw on the grid and label every line you draw; numbers in the boxes, working beside them.

A tax on backpacks: the line and the prices — three minutes alone

The grid shows the market for backpacks: demand D and supply S. The state puts a tax of 40 euros on every backpack sold; the sellers pay it to the state.



[5 points; drawing 2; boxes as marked]

(a) Draw the supply curve with the tax: at every quantity 40 euros above S. Label it S + tax.

(b) With the tax (quantity in thousand, prices in euros):

Quantity sold [1] | Buyers pay (euros a backpack) [1]

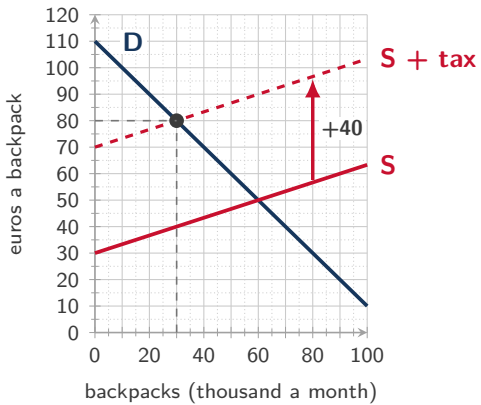
30 thousand	

Sellers keep (euros a backpack) [1]

Draw on the grid and label every line you draw; numbers in the boxes, working beside them.

A tax on backpacks: the line and the prices — three minutes alone

The grid shows the market for backpacks: demand D and supply S. The state puts a tax of 40 euros on every backpack sold; the sellers pay it to the state.



[5 points; drawing 2; boxes as marked]

(a) Draw the supply curve with the tax: at every quantity 40 euros above S. Label it S + tax.

(b) With the tax (quantity in thousand, prices in euros):

Quantity sold [1] | Buyers pay (euros a backpack) [1]

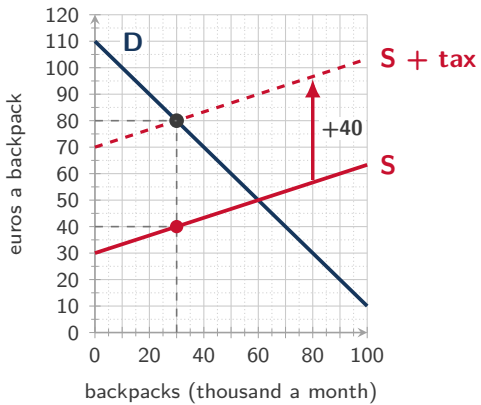
30 thousand	80 euros

Sellers keep (euros a backpack) [1]

Draw on the grid and label every line you draw; numbers in the boxes, working beside them.

A tax on backpacks: the line and the prices — three minutes alone

The grid shows the market for backpacks: demand D and supply S. The state puts a tax of 40 euros on every backpack sold; the sellers pay it to the state.



[5 points; drawing 2; boxes as marked]

(a) Draw the supply curve with the tax: at every quantity 40 euros above S. Label it S + tax.

(b) With the tax (quantity in thousand, prices in euros):

Quantity sold [1] | Buyers pay (euros a backpack) [1]

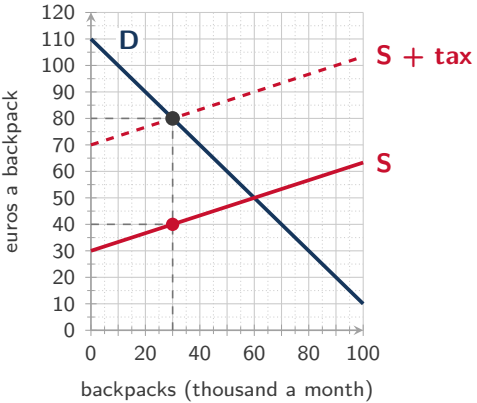
30 thousand	80 euros

Sellers keep (euros a backpack) [1]

80 − 40 = 40 euros

Draw on the grid and label every line you draw; numbers in the boxes, working beside them.

The tax's areas: numbers, and letters inside — five minutes alone



[9 points; letters 2; boxes as marked]

(c) With the tax of 40 euros a backpack, in thousand euros a month: CS, PS, T and DWL (below). Write each one's letters inside its area.

CS [1.5]	PS [1.5]	T [1.5]	DWL [1.5]

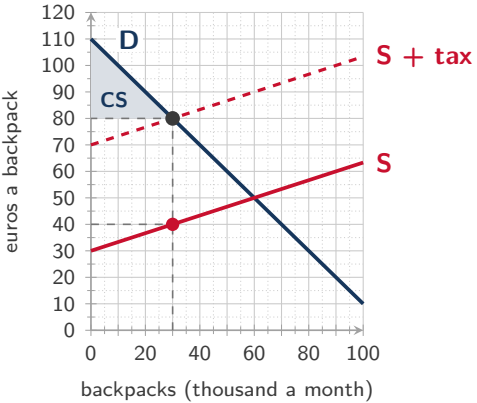
(d) Who bears more of the tax? Circle one. [1]

(Buyers) (Sellers)

(e) Same tax, but the buyers pay it to the state: which line would you draw? Would any box change?

CS: consumer surplus · PS: producer surplus · T: tax revenue · DWL: deadweight loss. An area's letters go inside it (a small area: beside it, with an arrow). A triangle is ½ × base × height.

The tax's areas: numbers, and letters inside — five minutes alone



[9 points; letters 2; boxes as marked]

(c) With the tax of 40 euros a backpack, in thousand euros a month: CS, PS, T and DWL (below). Write each one's letters inside its area.

CS [1.5]	PS [1.5]	T [1.5]	DWL [1.5]
CS = 450			

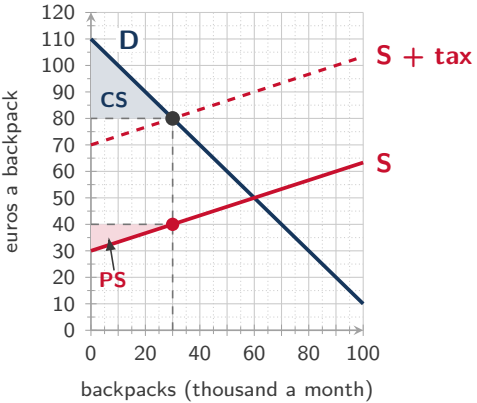
(d) Who bears more of the tax? Circle one. [1]

Buyers Sellers

(e) Same tax, but the buyers pay it to the state: which line would you draw? Would any box change?

CS: consumer surplus · PS: producer surplus · T: tax revenue · DWL: deadweight loss. An area's letters go inside it (a small area: beside it, with an arrow). A triangle is ½ × base × height.

The tax's areas: numbers, and letters inside — five minutes alone



[9 points; letters 2; boxes as marked]

(c) With the tax of 40 euros a backpack, in thousand euros a month: CS, PS, T and DWL (below). Write each one's letters inside its area.

CS [1.5]	PS [1.5]	T [1.5]	DWL [1.5]
CS = 450	PS = 150		

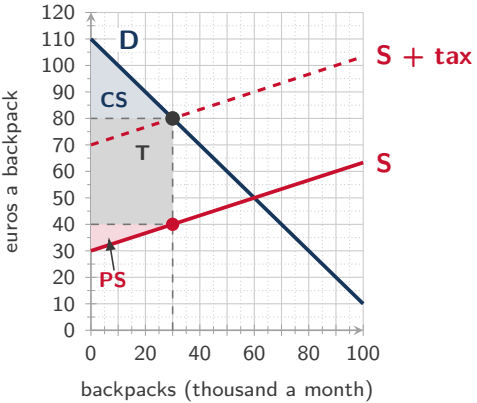
(d) Who bears more of the tax? Circle one. [1]

Buyers Sellers

(e) Same tax, but the buyers pay it to the state: which line would you draw? Would any box change?

CS: consumer surplus · PS: producer surplus · T: tax revenue · DWL: deadweight loss. An area's letters go inside it (a small area: beside it, with an arrow). A triangle is ½ × base × height.

The tax's areas: numbers, and letters inside — five minutes alone



[9 points; letters 2; boxes as marked]

(c) With the tax of 40 euros a backpack, in thousand euros a month: CS, PS, T and DWL (below). Write each one's letters inside its area.

CS [1.5]	PS [1.5]	T [1.5]	DWL [1.5]
CS = 450	PS = 150	T = 1,200	

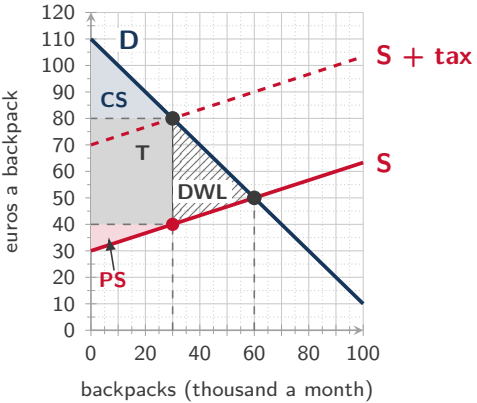
(d) Who bears more of the tax? Circle one. [1]

(Buyers) (Sellers)

(e) Same tax, but the buyers pay it to the state: which line would you draw? Would any box change?

CS: consumer surplus · PS: producer surplus · T: tax revenue · DWL: deadweight loss. An area's letters go inside it (a small area: beside it, with an arrow). A triangle is ½ × base × height.

The tax's areas: numbers, and letters inside — five minutes alone



[9 points; letters 2; boxes as marked]

(c) With the tax of 40 euros a backpack, in thousand euros a month: CS, PS, T and DWL (below). Write each one's letters inside its area.

CS [1.5]	PS [1.5]	T [1.5]	DWL [1.5]
CS = 450	PS = 150	T = 1,200	DWL = 600

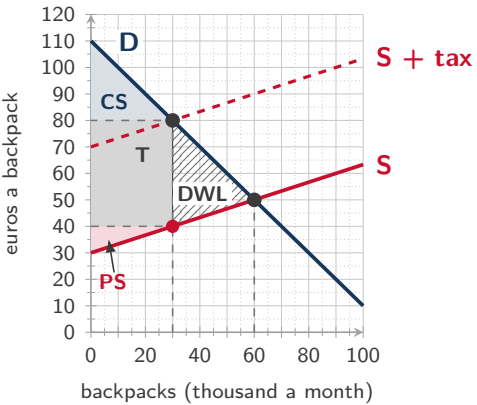
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(Buyers) (Sellers)

(e) Same tax, but the buyers pay it to the state: which line would you draw? Would any box change?

CS: consumer surplus · PS: producer surplus · T: tax revenue · DWL: deadweight loss. An area's letters go inside it (a small area: beside it, with an arrow). A triangle is ½ × base × height.

The tax's areas: numbers, and letters inside — five minutes alone



[9 points; letters 2; boxes as marked]

(c) With the tax of 40 euros a backpack, in thousand euros a month: CS, PS, T and DWL (below). Write each one's letters inside its area.

CS [1.5]	PS [1.5]	T [1.5]	DWL [1.5]
CS = 450	PS = 150	T = 1,200	DWL = 600

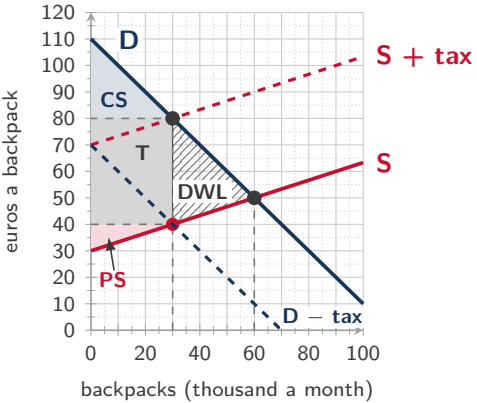
(d) Who bears more of the tax? Circle one. [1]

(Buyers) (Sellers) buyers pay 30 more (50 → 80), sellers keep 10 less (50 → 40)

(e) Same tax, but the buyers pay it to the state: which line would you draw? Would any box change?

CS: consumer surplus · PS: producer surplus · T: tax revenue · DWL: deadweight loss. An area's letters go inside it (a small area: beside it, with an arrow). A triangle is ½ × base × height.

The tax's areas: numbers, and letters inside — five minutes alone



[9 points; letters 2; boxes as marked]

(c) With the tax of 40 euros a backpack, in thousand euros a month: CS, PS, T and DWL (below). Write each one's letters inside its area.

CS [1.5]	PS [1.5]	T [1.5]	DWL [1.5]
CS = 450	**PS = 150**	**T = 1,200**	**DWL = 600**

(d) Who bears more of the tax? Circle one. [1]

(Buyers) (Sellers) buyers pay 30 more (50 → 80), sellers keep 10 less (50 → 40)

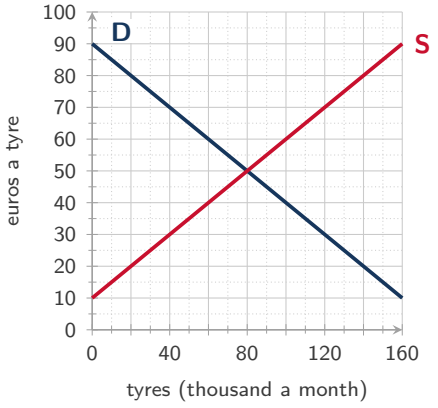
(e) Same tax, but the buyers pay it to the state: which line would you draw? Would any box change?

D − tax, 40 below D: it meets S at 30 thousand; sellers keep 40, buyers pay 80 a backpack — the same answers

CS: consumer surplus · PS: producer surplus · T: tax revenue · DWL: deadweight loss. An area's letters go inside it (a small area: beside it, with an arrow). A triangle is ½ × base × height.

Tyres: opening to trade, then a tariff — four minutes alone

The grid shows the market for tyres in a country that does not trade: demand D and supply S.



[6.5 points; drawing 2; boxes as marked]

(a) The country opens to trade at a world price of 30 euros a tyre. Draw it as a line; label it "world".

Production [1] Consumption [1] Imports [0.5]

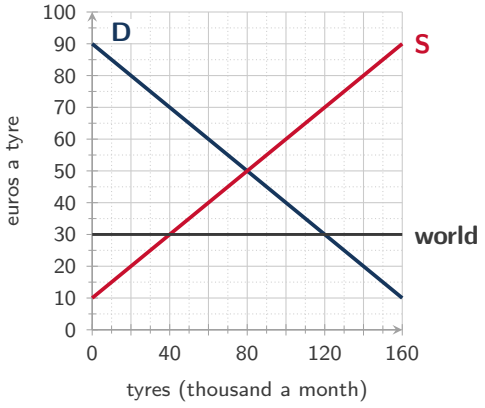
(b) Then a tariff of 10 euros on every imported tyre. Draw the price with the tariff; label it "tariff".

Price [0.5] Production [0.5]

Consumption [0.5] Imports [0.5]

Tyres: opening to trade, then a tariff — four minutes alone

The grid shows the market for tyres in a country that does not trade: demand D and supply S.



[6.5 points; drawing 2; boxes as marked]

(a) The country opens to trade at a world price of 30 euros a tyre. Draw it as a line; label it "world".

Production [1] Consumption [1] Imports [0.5]

(b) Then a tariff of 10 euros on every imported tyre. Draw the price with the tariff; label it "tariff".

Price [0.5] Production [0.5]

Consumption [0.5] Imports [0.5]

Tyres: opening to trade, then a tariff — four minutes alone

The grid shows the market for tyres in a country that does not trade: demand D and supply S.



[6.5 points; drawing 2; boxes as marked]

(a) The country opens to trade at a world price of 30 euros a tyre. Draw it as a line; label it "world".

Production [1] | Consumption [1] | Imports [0.5]

Production [1]	Consumption [1]	Imports [0.5]
40 thousand		

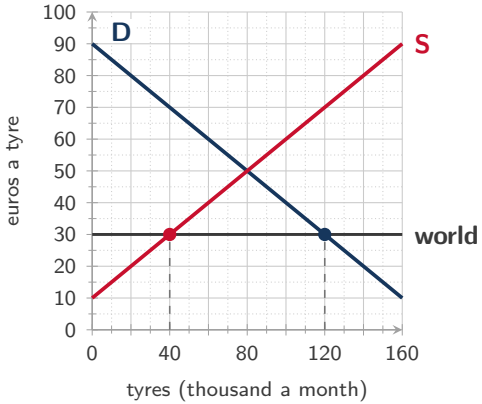
(b) Then a tariff of 10 euros on every imported tyre. Draw the price with the tariff; label it "tariff".

Price [0.5]	Production [0.5]

Consumption [0.5]	Imports [0.5]

Tyres: opening to trade, then a tariff — four minutes alone

The grid shows the market for tyres in a country that does not trade: demand D and supply S.



[6.5 points; drawing 2; boxes as marked]

(a) The country opens to trade at a world price of 30 euros a tyre. Draw it as a line; label it "world".

Production [1] | Consumption [1] | Imports [0.5]

Production [1]	Consumption [1]	Imports [0.5]
40 thousand	120 thousand	

(b) Then a tariff of 10 euros on every imported tyre. Draw the price with the tariff; label it "tariff".

Price [0.5] | Production [0.5]

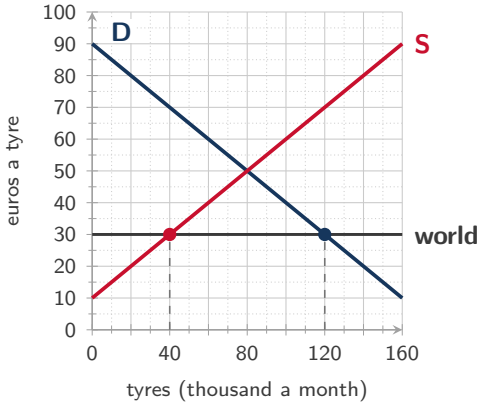
Price [0.5]	Production [0.5]

Consumption [0.5] | Imports [0.5]

Consumption [0.5]	Imports [0.5]

Tyres: opening to trade, then a tariff — four minutes alone

The grid shows the market for tyres in a country that does not trade: demand D and supply S.



[6.5 points; drawing 2; boxes as marked]

(a) The country opens to trade at a world price of 30 euros a tyre. Draw it as a line; label it "world".

Production [1]	Consumption [1]	Imports [0.5]
40 thousand	**120 thousand**	**120 − 40 = 80**

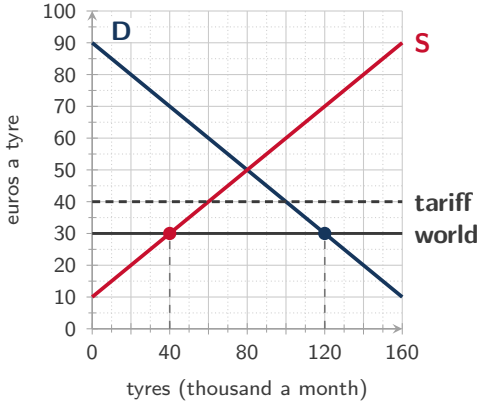
(b) Then a tariff of 10 euros on every imported tyre. Draw the price with the tariff; label it "tariff".

Price [0.5]	Production [0.5]

Consumption [0.5]	Imports [0.5]

Tyres: opening to trade, then a tariff — four minutes alone

The grid shows the market for tyres in a country that does not trade: demand D and supply S.



[6.5 points; drawing 2; boxes as marked]

(a) The country opens to trade at a world price of 30 euros a tyre. Draw it as a line; label it "world".

Production [1]	Consumption [1]	Imports [0.5]
40 thousand	**120 thousand**	**120 − 40 = 80**

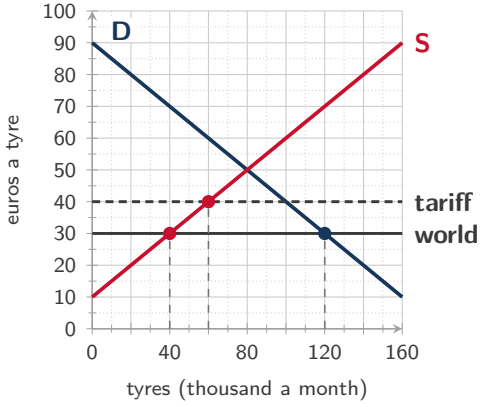
(b) Then a tariff of 10 euros on every imported tyre. Draw the price with the tariff; label it "tariff".

Price [0.5]	Production [0.5]
30 + 10 = 40 euros	

Consumption [0.5]	Imports [0.5]

Tyres: opening to trade, then a tariff — four minutes alone

The grid shows the market for tyres in a country that does not trade: demand D and supply S.



[6.5 points; drawing 2; boxes as marked]

(a) The country opens to trade at a world price of 30 euros a tyre. Draw it as a line; label it "world".

Production [1] | Consumption [1] | Imports [0.5]

Production [1]	Consumption [1]	Imports [0.5]
40 thousand	**120 thousand**	**120 − 40 = 80**

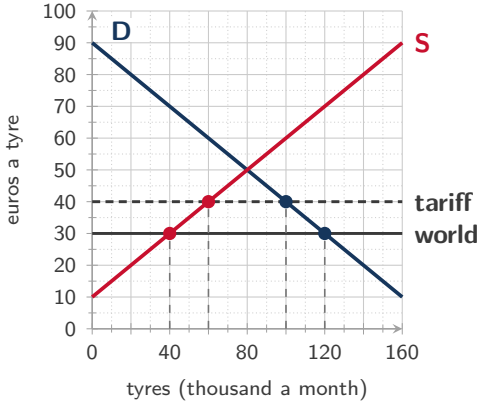
(b) Then a tariff of 10 euros on every imported tyre. Draw the price with the tariff; label it "tariff".

Price [0.5]	Production [0.5]
30 + 10 = 40 euros	**60 thousand**

Consumption [0.5]	Imports [0.5]

Tyres: opening to trade, then a tariff — four minutes alone

The grid shows the market for tyres in a country that does not trade: demand D and supply S.



[6.5 points; drawing 2; boxes as marked]

(a) The country opens to trade at a world price of 30 euros a tyre. Draw it as a line; label it "world".

Production [1]	Consumption [1]	Imports [0.5]
40 thousand	**120 thousand**	**120 − 40 = 80**

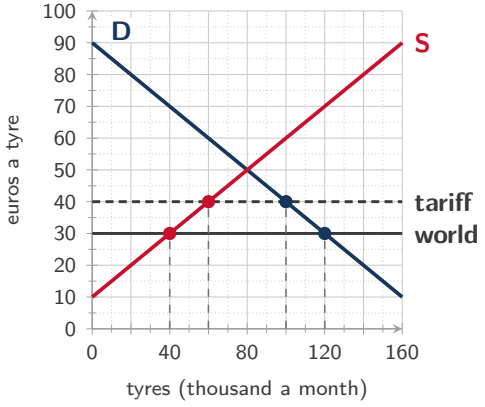
(b) Then a tariff of 10 euros on every imported tyre. Draw the price with the tariff; label it "tariff".

Price [0.5]	Production [0.5]
30 + 10 = 40 euros	**60 thousand**

Consumption [0.5]	Imports [0.5]
100 thousand	

Tyres: opening to trade, then a tariff — four minutes alone

The grid shows the market for tyres in a country that does not trade: demand D and supply S.



[6.5 points; drawing 2; boxes as marked]

(a) The country opens to trade at a world price of 30 euros a tyre. Draw it as a line; label it "world".

Production [1] | Consumption [1] | Imports [0.5]

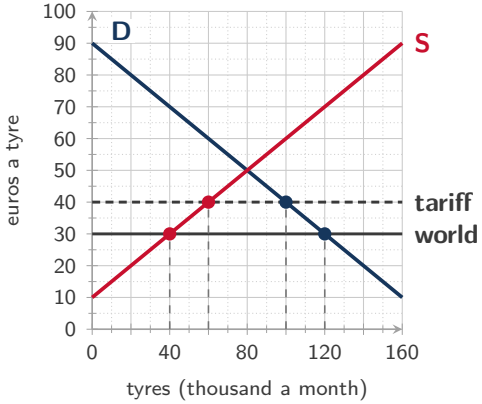
Production [1]	Consumption [1]	Imports [0.5]
40 thousand	**120 thousand**	**120 − 40 = 80**

(b) Then a tariff of 10 euros on every imported tyre. Draw the price with the tariff; label it "tariff".

Price [0.5]	Production [0.5]
30 + 10 = 40 euros	**60 thousand**

Consumption [0.5]	Imports [0.5]
100 thousand	**100 − 60 = 40**

The tariff's areas: T and two DWL triangles — three minutes alone



[5 points; letters 1; boxes as marked]

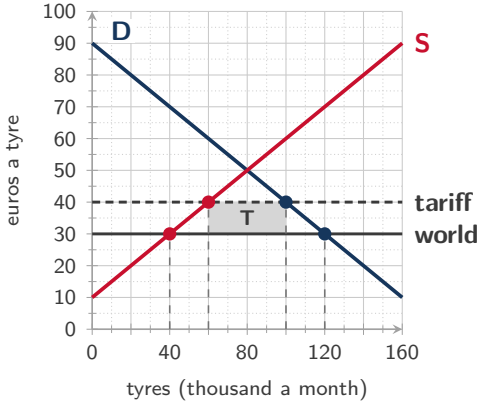
(c) With the tariff (world price 30, tariff 10), in thousand euros a month: the tariff revenue (T) and the deadweight loss (DWL, both triangles together). Write T inside its area, and DWL beside each of its two triangles, with an arrow.

T [2]

DWL (total) [2]

Thousand tyres × euros = thousand euros. A triangle is ½ × base × height.

The tariff's areas: T and two DWL triangles — three minutes alone



[5 points; letters 1; boxes as marked]

(c) With the tariff (world price 30, tariff 10), in thousand euros a month: the tariff revenue (T) and the deadweight loss (DWL, both triangles together). Write T inside its area, and DWL beside each of its two triangles, with an arrow.

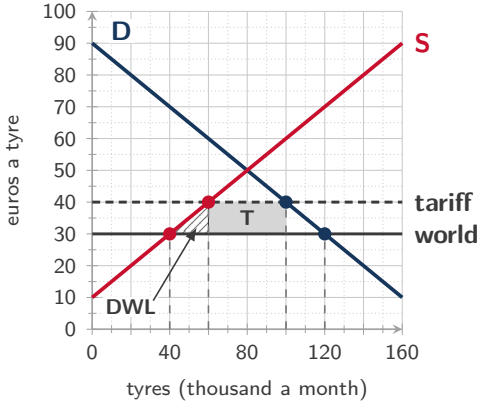
T [2]

T = 10 × 40 = 400

DWL (total) [2]

Thousand tyres × euros = thousand euros. A triangle is ½ × base × height.

The tariff's areas: T and two DWL triangles — three minutes alone



[5 points; letters 1; boxes as marked]

(c) With the tariff (world price 30, tariff 10), in thousand euros a month: the tariff revenue (T) and the deadweight loss (DWL, both triangles together). Write T inside its area, and DWL beside each of its two triangles, with an arrow.

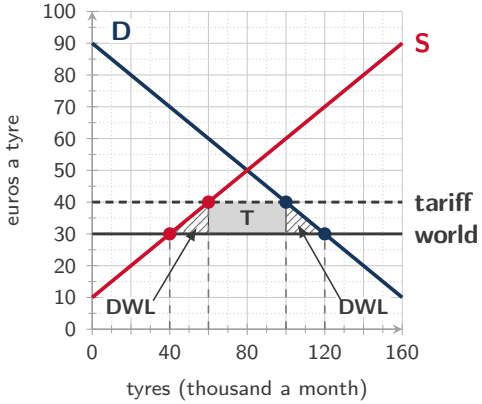
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The tariff's areas: T and two DWL triangles — three minutes alone



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T [2]

T = 10 × 40 = 400

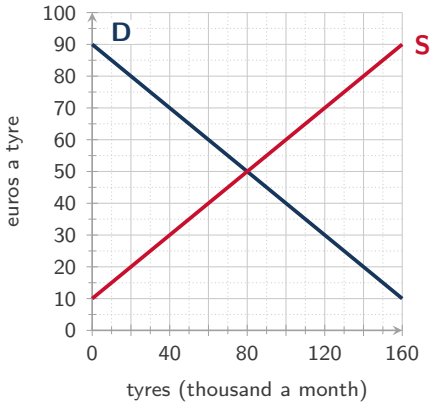
DWL (total) [2]

DWL = 100 + 100 = 200

Thousand tyres × euros = thousand euros. A triangle is ½ × base × height.

The same tyres with smoke: an external cost — four minutes alone

The tyre market without trade, D and S as before. Making each tyre also sends out smoke that harms people nearby: 30 euros of harm per tyre, which buyers and sellers do not count.



[10 points; drawing 4; boxes as marked]

(a) Draw the social-cost curve (SC: the sellers' cost plus the harm to others); label it SC.

(b) Market quantity [1] Efficient quantity [1.5]

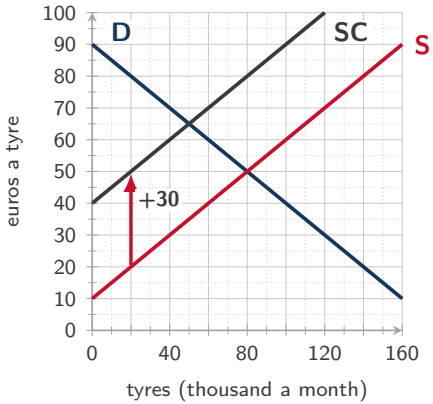
(c) The deadweight loss (DWL) at the market quantity, thousand euros a month; label its area DWL.

(d) The tax per tyre that brings the market to the efficient quantity.

(c) DWL [2] (d) Tax (euros a tyre) [1.5]

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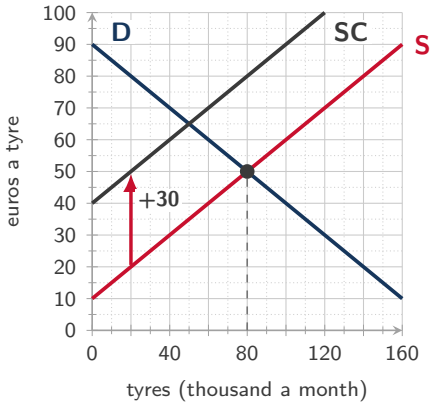
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[10 points; drawing 4; boxes as marked]

(a) Draw the social-cost curve (SC: the sellers' cost plus the harm to others); label it SC.

(b) Market quantity [1] Efficient quantity [1.5]

Market quantity	Efficient quantity
80 thousand	

(c) The deadweight loss (DWL) at the market quantity, thousand euros a month; label its area DWL.

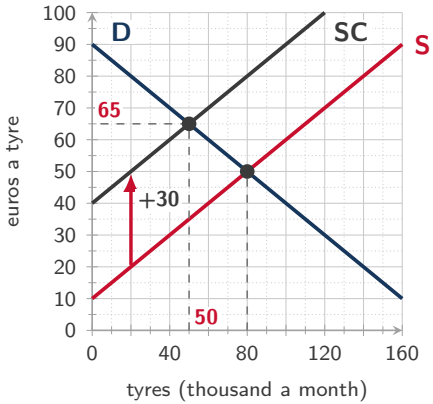
(d) The tax per tyre that brings the market to the efficient quantity.

(c) DWL [2] (d) Tax (euros a tyre) [1.5]

DWL	Tax (euros a tyre)

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[10 points; drawing 4; boxes as marked]

(a) Draw the social-cost curve (SC: the sellers' cost plus the harm to others); label it SC.

(b) Market quantity [1] Efficient quantity [1.5]

Market quantity	Efficient quantity
80 thousand	50 thousand

(c) The deadweight loss (DWL) at the market quantity, thousand euros a month; label its area DWL.

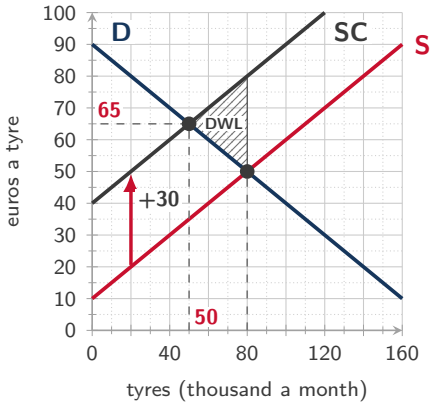
(d) The tax per tyre that brings the market to the efficient quantity.

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DWL	Tax (euros a tyre)

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The tyre market without trade, D and S as before. Making each tyre also sends out smoke that harms people nearby: 30 euros of harm per tyre, which buyers and sellers do not count.



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(a) Draw the social-cost curve (SC: the sellers' cost plus the harm to others); label it SC.

(b) Market quantity [1] Efficient quantity [1.5]

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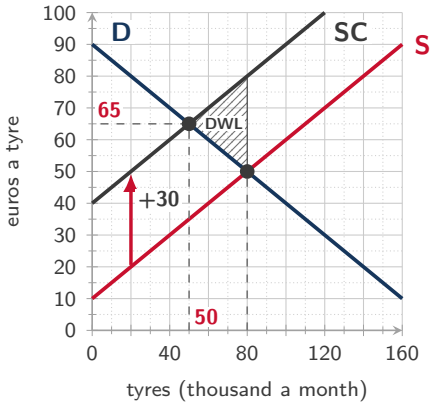
(d) The tax per tyre that brings the market to the efficient quantity.

(c) DWL [2] (d) Tax (euros a tyre) [1.5]

DWL = 450	

The same tyres with smoke: an external cost — four minutes alone

The tyre market without trade, D and S as before. Making each tyre also sends out smoke that harms people nearby: 30 euros of harm per tyre, which buyers and sellers do not count.



[10 points; drawing 4; boxes as marked]

(a) Draw the social-cost curve (SC: the sellers' cost plus the harm to others); label it SC.

(b) Market quantity [1] Efficient quantity [1.5]

Market quantity	Efficient quantity
80 thousand	50 thousand

(c) The deadweight loss (DWL) at the market quantity, thousand euros a month; label its area DWL.

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(c) DWL [2] (d) Tax (euros a tyre) [1.5]

DWL	Tax (euros a tyre)
DWL = 450	30 euros a tyre: the harm

Hoodies: from costs to a choice — four minutes alone (optional)

[8 points; boxes as marked]

A workshop sews hoodies and sells them at the market price. Its total cost, euros a week:

hoodies a week	0	25	50	75	100
total cost (euros)	400	700	900	1,300	1,900
marginal cost, MC (euros a hoodie)					

(a) Fill in the marginal cost (MC) of each extra 25 hoodies, per hoodie. [2]

(b) At 50 hoodies: the average variable cost (AVC)? The average total cost (ATC)? [2]

(c) The price is 20 euros a hoodie. How many hoodies? Profit? [2]

(d) The price falls to 14. Produce or shut down? Why? [2]

Variable cost = total cost − fixed cost (class 11, slides 9–11).
A price taker produces while MC < price (class 12, slide 8).

Hoodies: from costs to a choice — four minutes alone (optional)

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marginal cost, MC (euros a hoodie)		**12**	**8**	**16**	**24**

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AVC = (900 − 400) ÷ 50 = 10; ATC = 900 ÷ 50 = 18

(c) The price is 20 euros a hoodie. How many hoodies? Profit? [2]

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75 hoodies (MC 16 < 20 < 24): profit 1,500 − 1,300 = 200 euros

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75 hoodies (MC 16 < 20 < 24): profit 1,500 − 1,300 = 200 euros

(d) The price falls to 14. Produce or shut down? Why? [2]
produce 50: it loses 200 (700 − 900), less than the fixed cost of 400 it pays if it shuts (lowest price to produce: 10, the lowest AVC)

Variable cost = total cost − fixed cost (class 11, slides 9–11).
A price taker produces while MC < price (class 12, slide 8).

Before Thursday 12 November

- Thursday 12 November: monopolistic competition; problems on Blocks 3–4 — read Mankiw & Taylor Ch. 12, pp. 267–276
- Problem set 3: paper copy handed out today; bring your worked solutions and your copy on Thursday (not handed in)
- Practise in Sextant: the units the course page lists